

Arthur Labs Whitepaper

Watson Lewis-Rodriguez

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watson@arthurlabs.net

arthurlabs.net

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1 Disclaimer

Legal Disclaimer

This Whitepaper is published solely for informational purposes and to receive feedback and comments from the public. Nothing in this Whitepaper should be treated or read as a guarantee or promise of how Arthur Labs' business or its ecosystem will develop. This white paper outlines current plans, which could change at Arthur Labs' discretion, and the success of which will depend on many factors outside Arthur Labs' control, including market-based factors and factors within the technology and commerce industries, among others. Any statements about future events are based solely on Arthur Labs' analysis of the issues described in this white paper. That analysis may prove to be incorrect.

Finally, this Whitepaper is not finalized and is in continuous development. Future changes in references or additional content regarding infrastructural changes will be mentioned in the Whitepaper keynotes if they are major changes.

2 Abstract

Most people believe Arthur Labs is developing just another Real World Asset (RWA) blockchain. However, Arthur Labs is building a suite of systems, applications, and a blockchain with smart contract standards to facilitate the means of exchanging physical products, services, and deliveries.

Firstly, Arthur Labs is building out a suite of tools, which will be referred to as systems inside of this document. The first tool is called DEAN, a Digital Bazaar Factory. This has been shown to be the most successful to date. [SECTION]. There are additional tools explained in the same section.

Second, Arthur Labs is developing the Hyper Intelligent Innovation Engine (HIIE), an AI-driven multi-agent system that transforms raw human intent into physically buildable, ethically validated, and commercially viable engineered outputs. HIIE enables entrepreneurs worldwide to go from an idea to a patentable, manufacturable product — bridging the gap between human ambition and engineered reality. Where the marketplace systems (DEAN, ROSE) provide the storefront, HIIE provides the invention.

Third, Arthur Labs is building a dedicated blockchain for these systems, which aims to be a hallmark of blockchain innovation. This blockchain is named (The) VAULT. All tools provided by the labs are open source and accessible to developers and entrepreneurs in this blockchain.

A unique feature of this blockchain is its oracle validator model. This model allows for verification of the authenticity of the exchange, which provides the assurance of authentic agents inside the VAULT.

Fourth, Arthur Labs is publishing open-source standards of its smart contracts prepared as standards to be legally binding for its users and entrepreneurs.

Currently, Arthur Labs has developed out the commerce, service, and delivery contract factories that help developers and entrepreneurs deploy their businesses into Web3, reaping the advantages of authenticity and transparency that their corporate oligarchical competitors fail to provide.

Finally, Arthur Labs intends to provide a series of decentralized applications into its blockchain that helps abstract the information that blockchain fundamentally have such as account abstraction, tools, gas-less experiences, simple sign-on features, and FIAT payments.

With these five resources, Arthur Labs hopes to provide a future that Bitcoin, Ethereum and a vast variety of builders aim to create. These resources operate as a global black swan against corporate oligarchies, to enchant the quality of development for entrepreneurs and small businesses.

Recently, as Arthur Labs continues prioritizing self-sufficiency and tooling over external liquidity providers and market makers, there has been a recent development shift during 2025 to build out in conjoint partnerships to pursue new tooling, such as SUSAN, and to provide external website, application, and SEO building services for small businesses, called BUILDER.

3 Introduction

Corporate oligarchies and corrupt politicians maintain their grip through systems designed to conceal their actions from public scrutiny. Their growing fear of blockchain technology is warranted; transparency undermines the very mechanisms they rely on for control. As decentralized networks continue to evolve, these established powers face a simple truth: blockchain's immutable ledger makes opacity impossible. It is within this transformative moment that Arthur Labs emerges.

This document presents a series of tools and resources Arthur Labs will build to seamlessly onboard small startups onto blockchain technology to advance exchanges of physical goods and services. This vision at its core relies on transparency and trustless-ness to drive a series of new global exchange markets: Decentralized Commerce (DeCom), Decentralized Services (DeServ), and Decentralized Deliveries (DeDe).

Collectively, the 2025 market cap of the commerce, service and delivery markets are valued around \$25 Trillion. Yet for most small startups, trust and global outreach to potential clients are limited. Proposed in this document lays a series of plans and tools that help tackle these issues through automated and configurable systems and blockchain technology.

3.1 Tools and systems

Arthur Labs has developed a comprehensive suite of tools designed to revolutionize how entrepreneurs and developers build and scale digital marketplaces. At its core, these systems automate the creation of two-sided bazaars and marketplaces through configurable

components that can be deployed with either Web2 or Web3 functionality, depending on the builder's requirements.

Through a unified global onboarding form, any entrepreneur can initiate the marketplace generation process, selecting their preferred blockchain infrastructure or traditional web architecture. Each marketplace can be configured for commerce, services, or delivery operations, enabling rapid expansion into various market segments. Revenue generated from these marketplaces flows back into developing the VAULT blockchain's infrastructure, creating a sustainable ecosystem for future growth.

The system's primary advantage lies in its dramatic acceleration of development and deployment timelines. What traditionally takes 6-12 months can now be accomplished in days, significantly reducing the barrier to entry for new market participants. This efficiency is further enhanced by the QUINN System, which automatically generates and distributes marketing materials across multiple social media platforms, amplifying market reach without additional effort.

Looking ahead, these systems will integrate with Layer 2 applications to enable FIAT payment processing, creating a seamless bridge between traditional finance and blockchain technology. This hybrid approach ensures maximum accessibility while maintaining the benefits of decentralized systems.

The cornerstone of this ecosystem consists of several systems:

1. **DEAN System:** A digital bazaar factoryline, enabling rapid deployment of Web3-comptabile marketplaces, with full customization of components, smart contracts, and blockchain integration. Functional deployment across 7,500 different chains.
2. **ROSE System:** A parallel system focused on traditional Web2 marketplace development, providing a pathway for businesses preferring conventional infrastructure while maintaining upgrade potential to Web3.
3. **QUINN System:** An autonomous (and static) marketing tool that increases outreach for applications inside of the Arthur labs system, by seamlessly distributing content of marketplaces, or general business announcements onto multiple media platforms.
4. **SUSAN System:** An autonomous application building system using Model Context Providers, and a boilerplate format to full start-to-finish interactive apps. Clients inside SUSAN simply generate a business prompt, and SUSAN does a complete deep search analysis, reporting, and integration of the concept, from idea to MVP in three days.
5. **BUILDER System:** A nearly automated website builder tool that integrates beautiful design systems, APIs, SEO optimization, and various database tooling for clients around the United States.
6. **HIIE System:** The Hyper Intelligent Innovation Engine — a multi-agent AI system that transforms human intent into physically buildable, ethically validated, and

patent-ready engineered outputs. HIIIE produces PCB schematics, manufacturing line designs, bills of materials, provisional patent drafts, and full feasibility reports, enabling entrepreneurs to invent and manufacture products through Arthur Labs infrastructure.

Together, these systems form a comprehensive toolkit that democratizes both marketplace creation and physical product innovation, enabling entrepreneurs worldwide to rapidly iterate and launch their ideas while benefiting from blockchain technology's transparency and security. The following sections detail each system's architecture, capabilities, and implementation strategy.

3.2 AI-Driven Innovation

Arthur Labs recognizes that the primary bottleneck for global entrepreneurship is not access to markets — it is the translation layer between a human idea and an engineered, manufacturable product. Most entrepreneurs have concepts but lack the engineering organization, patent strategy, and manufacturing knowledge to bring them to physical reality.

The Hyper Intelligent Innovation Engine (HIIIE) eliminates this bottleneck. Developed under Arthur Labs in partnership with Registered Agentics (Marko Ruble), HIIIE operates as a complete AI-driven innovation pipeline: a team of specialized AI agents that takes a raw problem statement and produces validated engineering outputs ready for manufacturing and patent filing.

HIIIE's outputs feed directly into the Arthur Labs ecosystem. An entrepreneur can use HIIIE to design a physical product, deploy a marketplace for that product through DEAN or ROSE, register the exchange agreements on the VAULT blockchain through smart contract standards, and distribute marketing through QUINN — all within a single ecosystem. This creates an end-to-end pathway from invention to global commerce that no other platform currently provides.

Furthermore, HIIIE-generated designs and patent claims can be verified and timestamped through the VAULT's oracle validator model, providing immutable proof of invention origin and design authenticity. This positions the VAULT blockchain as a registry of AI-assisted innovation, where every engineered output carries a verifiable chain of custody from concept through validation to marketplace.

3.3 Blockchain

Arthur Labs is planning to build a standalone and independent blockchain for these new markets. By building an exceptionally low gas fee and fast processing network, various new marketplaces, bazaars and online stores can come into existence.

This aims to be a WASM and Solidity accepted blockchain, with various resources and boilerplate's to accelerate development time to test, debug and launch a product seamlessly.

Using a unique PoS validation mechanism with an oracle validator model that is incentivized to verify and publish the completion of transactions and real world exchanges.

Users can earn additional tokens for self verifying the completion of the service and publish this as receipts.

Theoretically, there is a way to include a juring and judging consensus mechanism at its base level to determine if a specific transaction is fraudulent. This mechanism will be discussed in the Layer 1 infrastructure section.

3.4 Contract standards

Above the blockchain lies a series of contract standards to ensure trade agreements are treated as legally enforceable. As of November 30th 2024, Arthur Labs has approximately a dozen smart contracts that are ready for testing and usage. These include multiple Factory Contracts (which generate contracts through tailored user inputs), for selling goods, purchasing goods, and payment proxy agreements when sellers fulfill provide physical delivery receipts for their commerce. The remaining include RWG contracts, RWS contracts, and RWD contracts. As Arthur Labs continue its development for these contracts, these boilerplates will remain open source and publicly accessible in Solidity and Rust through authorized organizations GitHub repositories.

3.5 Layered applications

Going beyond the contract standards are the built-in and free to use (excluding gas fees + various transaction/exchange fees) dApps. The following is just a few applications planned for development:

1. **Wallet Provider:** The essential tool for consumers and token holders to be able to interact in Arthur Labs.
2. **Account Abstraction:** This stores and provides consumers an easy to access wallet connection (both functional inside of the application, and on the Wallet Provider), node providers inside will received a sharded piece of the wallet, so the accounts are held encrypted on chain in fragments so that no other user or provider can access the consumers wallet.
3. **Fiat to token bridge:** Arthur Labs additionally will provide a liquidity pool that lets user pay in their native currency, and the pool will facilitate the on chain transaction. Interest or transaction fees accrued will be given to the staking facilitators.
4. **Gas providers:** Some dApps across Arthur Labs may be interested in using a tool in which the marketplace, or platform covers the transaction fees on behalf of the user for a more seamless experience. This tool provides such, being able to cover the gas fee on behalf of the user up to entire cost of the transaction.

4 Architecture

4.1 Layer 1 infrastructure

The VAULT is deployed as a standalone Proof of Stake blockchain. At the time of v1.0.6 of this whitepaper, the primary focus has building the DEAN System, however with the system launch, revenue generated will fund the development process of the VAULT.

4.1.1 Core functionality

The blockchain handles all applications built onto the VAULT. Each block holds dynamic and customized information about real-world activities, backed by oracle validators. Each block hosts typical transfers, as well as stores a list of oracles that act as the confirmation of a specific transaction. Oracles are users who manually review a transaction or authenticate a physical exchange, whether they prove delivery receipts in commerce marketplaces, or verify completions of physical services, for example.

The core accepted programming languages for contract deployment will be Solidity, and WASM.

4.1.2 Oracle validation

The oracle validator model allows each user to incentivize authenticity exchange of a product, service or delivery distributed by contractually paying a third party individual to complete the service of validation of a real-world activity. For example:

- paying an automated node validator to authenticate a physical receipt of a product delivery
- having a validator determines the quality and authenticity of a real world service and its resulting outcome
- having a person view a completion of a construction of property or result of a photography service.
- having a worker at a logistics company confirm with a photo that a CDL truck driver is at the correct delivery location.
- signing a contract of fulfillment when delivering freight.

4.1.3 Data security

Zero-knowledge proof is optionally implemented for smart contract owners or users of an Arthur Labs bazaar, or optionally encrypting data through salt rounds. An oracle validator will need to be able to request proof of information upon request.

The VAULT also has a mission to be on the frontline for encryption methods. Arthur Labs aims to partner with various quantum hacking prevention organizations like Quantum EVM.

4.1.4 Proof of Stake System

The VAULT implements a robust Proof of Stake (PoS) consensus mechanism that serves both transaction validation and oracle validation purposes. The native staking currency is VAU, which governs blockchain transactions and enables staked nodes to work as standalone operators.

The validation process combines traditional PoS mechanics with specialized oracle validation capabilities:

Transaction Validation: Stakers can participate in:

- Block validation and consensus
- Fee earning from transaction processing
- Network security maintenance

Oracle Validation Integration: The PoS system uniquely integrates with the oracle validation framework, allowing stakers to:

- Validate physical deliveries and real-world services
- Earn additional rewards for oracle services
- Participate in multi-signature validation for high-value transactions

The economic model follows:

$$\text{Total Rewards} = \text{Base Staking Reward} + \text{Transaction Fees} + \text{Oracle Service Fees} \quad (1)$$

Security measures include minimum staking requirements, slashing conditions, and time-locked withdrawals to ensure network integrity.

4.1.5 Native Currency

The native currency of the VAULT is VAU, which serves as the primary medium of exchange across the ecosystem.

Core Utility Functions VAU serves multiple essential purposes within the ecosystem:

1. **Primary Transaction Currency:**

All VAULT transactions utilize VAU as the primary medium of exchange. This includes marketplace purchases, service payments, and cross-chain operations.

2. **Gas Fee Mechanism:**

Network operations require VAU for gas fees, ensuring efficient resource allocation and spam prevention. The gas fee structure is dynamically adjusted based on network utilization.

3. **Staking Operations:**

VAU serves as collateral for network validation, with stakers receiving rewards proportional to their contribution. This creates a secure and decentralized consensus mechanism.

4.1.6 Technical Foundation

4.2 Layer 2 Infrastructure

4.2.1 Technical Foundation

Layer 1 is developed with EVM compatibility and built for WASM applications and Rust contracts.

4.2.2 Core Functionality

Layer 2 serves as the scaling solution that issues all of the smart contract standards that can be considered bilateral and legally binding to the participants of a transaction. It provides dynamic contract input data that allows for anyone to customize a specific smart contract transaction.

4.2.3 Contract Standards

It is important to mention the use of factory contracts wrapped around the following contract standards. These are best used when deploying a bazaar or marketplace. Factory contracts essentially provide all listing information and automatically fills out the listing information into a new deployed contract for the listing provider. Because this automates the generation of new contracts, these are most efficient inside public platforms where people can list and sell related items.

However, when one entrepreneur or developer is listing their item information in a dedicated website, it is best to deploy a series of customized contracts so that no one can self generate contracts based on the factory format.

Factory contracts store an array of listings generated from the factory, and if is used inside of permissioned websites where there is a limited amount of items, it may show array problems if the website maps out every item.

The standards proposed for the initial infrastructure consists of:

1. Physical Product Contracts This is an upgradable smart contract standard that allows for any seller of a physical product to share the product information, price, quantity, delivery methods, delivery costs, and more. This contract needs to be upgradable because it allows for anyone to make a new contract by purchasing the information displayed in the contract, as well as modifying the new contract information, such as the quantity purchased, customer's delivery address, customer's name, and more. Importantly, the customer metadata must be converted and encrypted effectively, complying with GDPR and general data protection ethics - this can be encrypted using salt rounds from the private signing key, which is only decryptable by a given list of permissioned users found in the prime smart contract standard. Using zero-knowledge proofs to handle the encryption and decryption of consumer data may be the most valuable output.

2. Physical Service Contracts Similar in the sense to the physical product contract standards, this is an upgradable smart contract standard that allows for any service

provider to issue out promises of physical activities for reward of compensation. Service providers need to provide the tasks they provide, location it is provided, and the results necessary to receive the prize. Customer data also needs to be encrypted similarly by upgrading a new smart contract which is only accessible to the contract owner and its list of permissioned users.

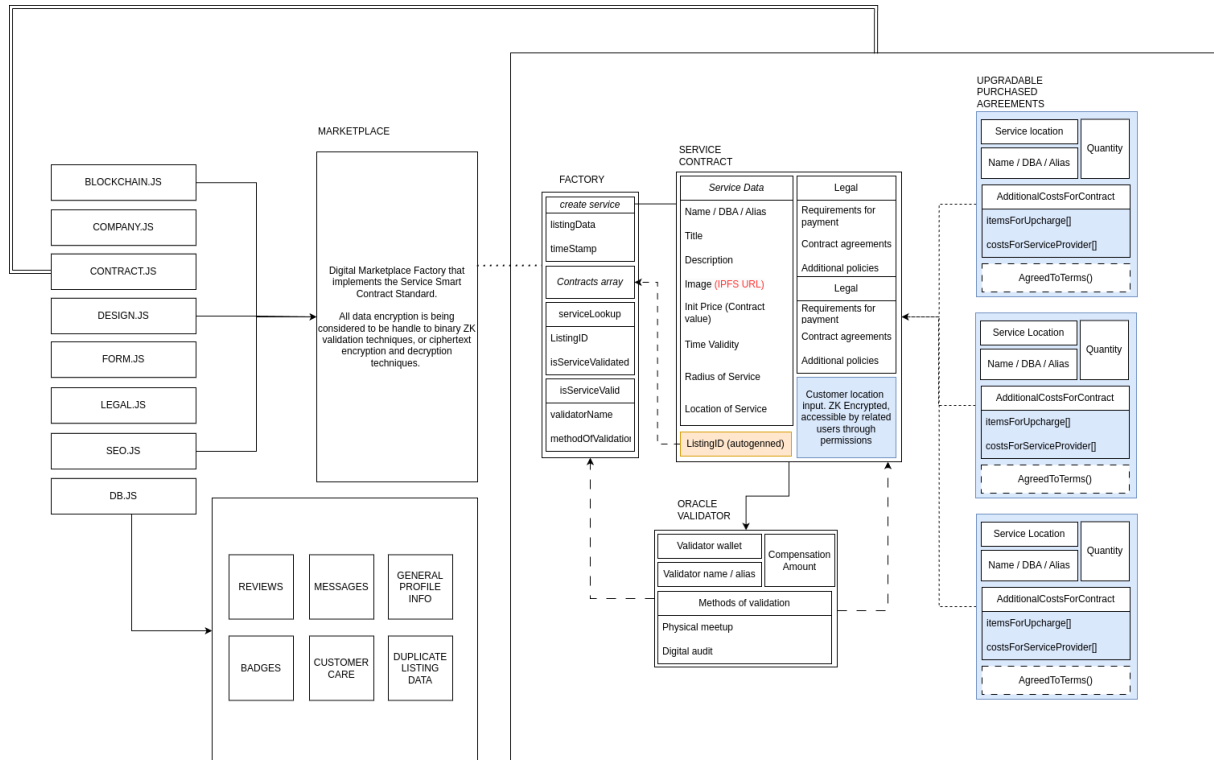


Figure 1: The Service smart contract infrastructure, including marketplace implementation. There is not currently a demo application pulic.

3. Delivery Service Contracts This is the smart contract standard that allows for any delivery driver around the world to provide and distribute service of delivery of a given product or item in exchange for compensation. This smart contract provides a few unique features as well that is optional depending on the contract creator. For instance, being able to share and upload geolocation is essential for a lot of couriers being able to handle freight, being able to constantly update the initial standard allows for digital proof and timestamping of a physical delivery. Additionally, this is an upgradable smart contract in another sense that customers interested in purchasing delivery contracts need to provide customized instructions, pickup data, dropoff data, and be accessible for potential black swan events.

4.3 Layer 3 infrastructure

Arthur Labs will be developing out required and functional applications to help accelerate development, as well as provide necessary tooling to reach the traditional consumer market.

A few of these applications will dedicate its efforts into making as much of a traditional system as possible, where clients can process and checkout payments using traditional FIAT currencies to purchasing items, while benefiting from the enchanting effects of blockchains.

The majority of applications built from Arthur Labs into the VAULT include governance and voting platforms for the DAO, injected wallet providers, transaction judging and moderation, and a variety of mentioned applications that offer unique features.

4.4 Layer 4 Infrastructure

4.4.1 Consumer protection

Standalone business models can be built in the Arthur Labs ecosystem to handle consumer failures. For instance, being able to moat an insurance model around decentralized applications provides assurance for consumers that they can safely and seamlessly use the application and be compensated in case the blockchain isn't efficient.

4.4.2 Payment Infrastructure

Additional infrastructure such as account abstraction and guest accounts allow for users of the Arthur Labs ecosystem to not even need to touch blockchains in order to reap its benefits. Similar to Coinbase Guest, wallet providers in Arthur Labs may provide a debit or credit card payment method that entirely removes the need for users to interact with cryptocurrencies. By allowing the user to pay a weekly maximum amount (\$500) without needing to own a specific account, it allows for users to continue to browse and utilize the Arthur Labs ecosystem without even knowing.

4.4.3 User Account Management

Similarly to guest accounts for payments, users may also register through Layer 2 infrastructure using emails and passwords, which then automatically provides the user a unique wallet. Advanced and blockchain-savvy users can easily go through the settings in various protocols to easily change their account.

5 Provided Systems

5.1 DEAN System

DEAN is a digital marketplace factory for building two-sided marketplaces in Web3 compatible chains. Throughout or initial research phase in a previous Decentralized Courier Protocol (DCP), Arthur Labs found that most generalized marketplaces contain approximately 25-30 boilerplate components necessary for any marketplace to function, including explore sections, checkout processes, messaging features, profiles, and other various UI components. DEAN essentially builds out a dynamic and variable based approach for building any RWG, RWS or RWD marketplace by streamlining the component implementation, database connection, and other configurations essential to launching a marketplace.

Arthur Labs first built out the DEAN system prior to the ROSE system, as DEAN can

As ROSE develops further into a BYO MaaS, there is an intent to partner with ChaiBuilder (<https://chaibuilder.com/>) to build an interactive platform that imports the components from DEAN and configuration system to automate marketplace development for the creator.

Long term, both ROSE and DEAN aims to build a Lego-like architecture so any developer or builder can access the platform and deploy a new marketplace concept quickly, while Arthur Labs receives percentage ownership and revenue from each transaction.

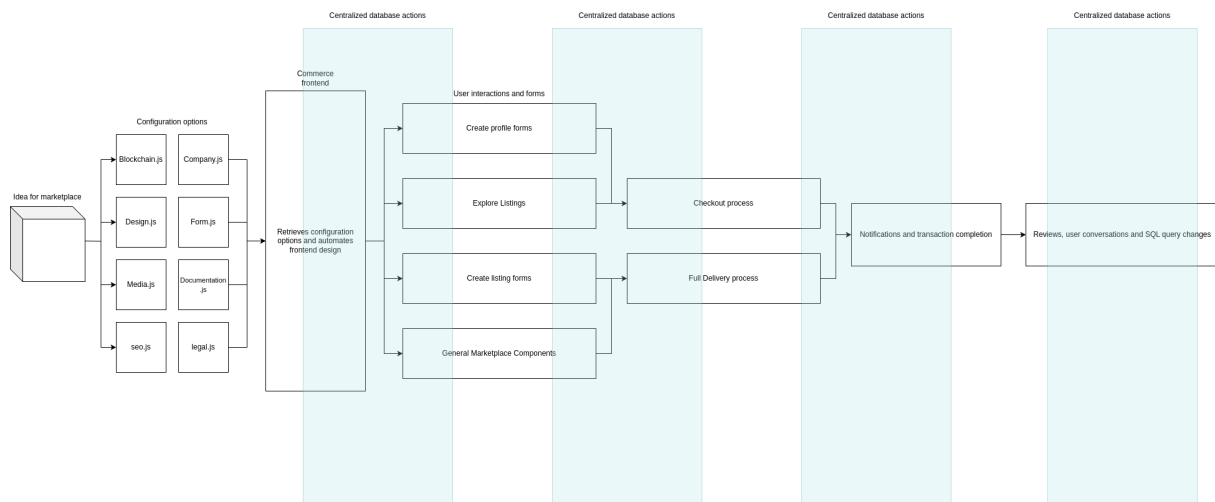


Figure 3: The System architecture for ROSEs commerce marketplace. There is not currently a demo application public.

Seen above, this system stores all data in a centralized database and processes payments through a traditional payment gateway API. Idealistically, Arthur Labs runs a few regional marketplaces that help generate revenue for the DAO to fund ongoing ecosystem and marketplace development.

5.3 QUINN System

QUINN is a deployable feature for any Arthur Labs marketplace, in which is a form replication system that refactors and formats marketplace listings into various social media accounts.

This ultimately streamlines marketing costs for small businesses so that they can easily share and post their listings through a list of professionally designed media cards, in which once the business owner connects their business accounts (Instagram, Meta, X, Discord, Telegram, Element, Facaster, and any of social platform that takes in a similar set of form inputs)

The goal with implementing QUINN with the marketplaces is so entrepreneurs forming their own businesses can increase their media exposure by over 500% while reducing the time for manual replication by 80%.

This system is permanently free in the ecosystem but may be optionally licensed out to

other blockchain networks.

5.4 SUSAN System

SUSAN is a conjoint effort from three companies, including Satoshi-O's, a serial startup agency in Texas, and Frank Bevr, a contractor based in Germany, and Arthur Labs. SUSAN is a comprehensive application and tooling system that follows a sequential list of tasks to build a thorough application.

Firstly, SUSAN asks a client what business concept they have, and if that is under an existing company, or to tailor its designs and code from scratch. After receiving the clients input, and various design references, the tool completes a deep research report about the business opportunity, and frameworks it should use. Once the report is concise and completed, it proceeds to generate a new repository for its code, likely using NextJS and Tailwind CSS.

With a series of already integrated Model Context Providers (MCP), SUSAN proceeds to build out an application tailored to the design references and report analysis. Initially the application builds out the design schematics, then proceeds to build functions and necessary components to build an MVP. Once these sections are near completed, SUSAN continues on with MCP functionality by reading the clients API connections to various payment providers, database handlers, and other necessary APIs to build out the MVP. The MCP connection retrieves existing structures, and modifies the clients connections to fit the functionality of the application. After its connections and completion, SUSAN then handles SEO optimization and various build changes. Finally, one of the three companies overviews and finalizes any core design changes or building changes and manually modifies the program to ensure proper functionality ready for MVP and production.

5.5 BUILDER System

BUILDER is an Arthur Labs website builder, dedicated primarily to contract development work to various USA sites, but is globally accessible for work. Revenue generated from BUILDER proceeds to fund the core development of the overall ecosystem.

5.6 HIIE System

The Hyper Intelligent Innovation Engine (HIIE) is a novel class of AI system developed under Arthur Labs in partnership with Registered Agentics (<https://registeredagentics.ai/>). HIIE transforms raw human intent into physically realizable, ethically validated, and commercially viable engineered outputs. It is not a chatbot or code assistant — it is a complete innovation pipeline that takes the rawest form of human ambition and, through recursive reasoning, multi-agent collaboration, and physics-grounded validation, produces outputs that can enter the physical world.

5.6.1 Multi-Agent Architecture

Each HIIE project spawns a full team of specialized AI agents, each with a distinct engineering role:

- **Theoretical Engineer** — first-principles reasoning and conceptual design across hardware, materials, and manufacturing domains
- **Materials Engineer** — material selection optimized for US-accessible sourcing, cost, and environmental lifecycle
- **Manufacturing Process Engineer** — factory line design, batch optimization, yield modeling, and quality control
- **Research Group** — three concurrent agents performing live retrieval from patent databases (USPTO, EPO, WIPO), academic papers, and standards bodies
- **Patent Strategist** — real-time global patent landscape monitoring, novelty window identification, and provisional claim drafting
- **Feasibility Manager** — scores all outputs against technical, manufacturing, commercial, and timeline feasibility thresholds
- **Ethics Officer** — reviews for dual-use risk, environmental impact, and societal consequences with immutable audit logging
- **Project Manager** — coordinates all agents, tracks milestones, and manages resource allocation

A persistent AI co-director named Arthur maintains long-term project memory across sessions, flags output drift, and advocates for the user’s original intent throughout the process.

5.6.2 Anti-Slop Validation

HIIE treats the prevention of AI hallucination as an architectural requirement. Every output undergoes three layers of validation before delivery:

1. **Recursive Validation with Divergence Enforcement** — outputs are iteratively self-critiqued and cross-challenged between agents. When outputs converge too closely (cosine similarity ≥ 0.92), the system forces divergent reasoning to prevent circular reinforcement of errors.
2. **Human Perspective Injection** — at defined checkpoints, HIIE generates both a layperson summary and domain-expert critique of every technical output. A feasibility score is computed across technical, physical, commercial, and ethical dimensions. Outputs scoring below threshold are rejected.
3. **Physical Ground-Truth Anchoring** — material properties are sourced from NIST and ASM International databases, electrical designs are validated against PySpice simulation, mechanical designs are stress-tested in FreeCAD, and chemistry outputs are cross-referenced against PubChem before feasibility is asserted.

5.6.3 Ethics Governance

Ethics is embedded as a continuous layer throughout HIIIE's pipeline, not a terminal filter. Six core pillars govern all outputs: harm prevention, environmental responsibility, full transparency, human primacy, equitable design, and an immutable audit trail. HIIIE cannot self-modify its ethics framework — this layer is protected from all instruction overrides. Designs for weapons systems, biological or chemical weapons, mass surveillance, or any system whose primary purpose is to cause physical harm are architecture-level constraints that cannot be overridden by any user or instruction.

5.6.4 Output Types

HIIIE produces complete engineering deliverable packages:

- Technical specification sheets with dimensions, tolerances, and material callouts
- Bills of materials with US sourcing, costs, and live supplier availability
- Provisional patent application drafts with claims, abstracts, and technical drawings
- Schematic and CAD files (KiCad, FreeCAD, SVG) auto-generated by domain agents
- Viability reports with feasibility scores, cost estimates, and risk matrices
- Environmental impact reports with carbon footprint analysis and net-positive pathways
- Manufacturing process flow diagrams and factory line designs

5.6.5 Reference Use Cases

HIIIE's intended scope spans precision instrumentation to large-scale manufacturing:

1. **Localized Atomic Force Microscope** — a cantilever-based AFM within a 3×3 ft envelope with remote WebRTC control, including full mechanical CAD, PCB schematic, BOM, and provisional patent draft.
2. **Novel vRAM Architecture** — SRAM-on-DRAM stacking alternatives using domestically available materials, targeting bandwidth > 1 TB/s at < 30% of current HBM2e material cost.
3. **GPU Batch Manufacturing System** — full factory line design including wafer handling, die bonding, burn-in testing, yield optimization, and facility layout CAD.
4. **Environmentally Net-Positive AI Compute Machine** — hardware specification targeting a positive carbon offset through waste heat capture, on-site solar generation, and renewable integration.

5.6.6 Integration with Arthur Labs Ecosystem

HIIE's outputs connect directly into the broader Arthur Labs infrastructure:

- **DEAN / ROSE Integration** — entrepreneurs use HIIE to design a physical product, then deploy a marketplace for that product through DEAN or ROSE, creating a complete idea-to-storefront pipeline.
- **VAULT Blockchain Registration** — HIIE-generated designs, patent claims, and engineering outputs can be timestamped and verified through the VAULT's oracle validator model, providing immutable proof of invention origin and design authenticity.
- **Smart Contract Standards** — the physical goods produced from HIIE designs are governed by RWG contract standards on the VAULT, ensuring legally enforceable trade agreements for AI-engineered products.
- **QUINN Marketing** — products invented through HIIE and listed on DEAN/ROSE marketplaces flow into QUINN for automated multi-platform marketing distribution.

5.6.7 Infrastructure and Deployment

HIIE Phase 1 runs on dedicated Mac Mini M4 Pro nodes hosted at Registered Agencies facilities. Apple Silicon's unified memory architecture eliminates the PCIe bandwidth bottleneck present in traditional configurations, enabling efficient multi-agent inference on a single node. Builder and Enterprise subscribers receive a dedicated machine that accumulates trained specialist model adapters over the contract term — the hardware asset appreciates through accumulated project intelligence. A detailed technical specification of HIIE's architecture, agent delegation framework, resource economy, and fine-tuning methodology is available in the companion HIIE Technical Whitepaper.

5.7 Platform Revenue Model

Every marketplace produced in both DEAN and ROSE is aimed to have a targetted marketplace owner in a standalone EVM compatible chain. Additionally, marketplaces are produced to only be viable regionally. Ideally, Arthur Labs hosts and maintains a series of platforms across leading chains, like Ethereum, Solana, Sui, Rootstock, Polkadot, and Cardano.

Revenue generated from the centrally hosted chains will be invested into building the larger ecosystem.

Applicants are encouraged to apply for a customized marketplace through the <https://arthurlabs.net/> application form. Once approved, applicants will receive a list of offers for split revenue ownership of the marketplace.

All marketplaces are hosted in a central code branch, where general marketplace bugs and development can cross develop, saving overall costs. All marketplaces are additionally encouraged to branch and develop out unique features and components necessary to

individualize their marketplace, but use the DEAN factory as an initial boilerplate to produce a USP.

Generally speaking, Arthur Labs will aim to receive 2.5% transaction fees for product sell contracts to be generated into the marketplace + a flat \$0.25. Then an additional 2.5% and \$0.25 is charged to the contract purchaser, with a final 2.5% and \$0.25 final charge for both parties during the payment proxy stage.

For instance, a \$20.00 item will earn a total of 10% in transaction fees, and a \$1.00 flat fee, as both parties pay 5% and \$.50. Earning a 15% margin.

5.7.1 HIIE Innovation Engine Revenue

In addition to marketplace transaction fees, the HIIE system introduces a subscription-based revenue stream for AI-driven invention services:

Tier	Price	Includes
Explorer	\$299/month	5 projects/month, core specifications + BOM + viability report
Builder	\$999/month	20 projects/month, full output package + patent draft + dedicated Mac Mini at Registered Agentics
Enterprise	Custom	Unlimited projects, dedicated agent configuration, white-label output, SLA, Registered Agentics hardware contract
Developer API	\$0.05/token	Pay-per-use pipeline access for third-party integrations

Builder and Enterprise subscribers receive a dedicated Mac Mini M4 Pro at Registered Agentics facilities. After the contract year-end, the machine is physically transferred to the client — carrying not just the hardware, but twelve months of trained specialist model adapters representing accumulated project intelligence. Revenue from HIIE subscriptions is reinvested into VAULT blockchain development, HIIE model improvements, and ecosystem expansion.

6 Additional Notes

6.1 Initial Marketplace Deployments

As of January 2025, Arthur Labs is nearing completion of the demo application with full Web3 functionality. The strategic deployment across multiple blockchain ecosystems maximizes accessibility while leveraging the unique strengths of each network. Arthur Labs will launch in the following ecosystems:

6.1.1 Polygon

<https://polygon.technology/> Polygon's ecosystem stands out for its commitment to zero-knowledge cryptography and cost-efficient transactions, making it ideal for high-frequency marketplace activities. Their extensive developer tooling and robust infrastruc-

ture provide Arthur Labs with the necessary framework to implement advanced privacy features and scalable solutions. The network's dedication to ZK technology aligns perfectly with Arthur Labs' vision for secure, private consumer data handling in decentralized commerce.

6.1.2 Optimism

optimism.io/ Optimism's innovative approach to Layer 2 scaling through optimistic rollups offers Arthur Labs the ability to process large volumes of marketplace transactions with minimal fees. Their established presence in the DeFi ecosystem provides a ready audience for real-world commerce applications. The platform's dedication to public goods funding also aligns with Arthur Labs' mission to democratize marketplace access. Currently, Arthur Labs utilizes the opBNB Testnet for development and testing, demonstrating the viability of the platform's architecture.

6.1.3 Ethereum

ethereum.org As the foundation of decentralized applications, Ethereum provides Arthur Labs with unparalleled security and network effects. The ecosystem's maturity and widespread adoption make it essential for establishing credibility and accessing the largest possible user base for marketplace deployment. Ethereum's upcoming developments in scaling and efficiency will further enhance marketplace functionality.

6.1.4 Arbitrum

<https://arbitrum.io/> Arbitrum's Layer 2 solution offers Arthur Labs the perfect balance of Ethereum's security with improved throughput and reduced costs. The platform's compatibility with existing Ethereum tools and infrastructure allows for seamless deployment of marketplace contracts while maintaining high performance and low transaction fees. This makes it particularly suitable for marketplaces with high transaction volumes.

6.1.5 Rootstock

<https://rootstock.io/> Rootstock's unique position as a Bitcoin sidechain with smart contract capabilities opens up unprecedented opportunities for Arthur Labs. As a leader in Bitcoin-based EVM deployments, Rootstock provides access to the Bitcoin ecosystem's massive user base and liquidity. This integration is particularly valuable for:

- **Global Reach:** Enabling marketplace deployment in regions where Bitcoin adoption is high, particularly in developing economies.
- **Bitcoin Integration:** Leveraging Bitcoin's network effect while providing smart contract functionality.
- **Regional Markets:** Facilitating localized marketplace creation in crypto-forward jurisdictions like El Salvador and Liberland.
- **Financial Inclusion:** Providing entrepreneurial tools to regions with limited access to traditional financial infrastructure.

This multi-chain deployment strategy ensures that Arthur Labs can serve diverse market needs while leveraging each network's unique advantages. By maintaining presence across these key ecosystems, we create a robust and accessible platform for global commerce, enabling entrepreneurs worldwide to participate in the decentralized economy regardless of their technical expertise or geographic location.

References

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